

**Actrapid®**

Penfill®

100 IU/ml

Solution for injection in cartridge.

**Qualitative and quantitative composition**Insulin human, rDNA (produced by recombinant DNA technology in *Saccharomyces cerevisiae*).

1 ml contains 100 IU of insulin human.

1 cartridge contains 3 ml equivalent to 300 IU.

One IU (International Unit) corresponds to 0.035 mg of anhydrous human insulin.

**Pharmaceutical form**

Solution for injection in cartridge.

Clear, colourless, aqueous solution.

**Therapeutic indications**

Treatment of diabetes mellitus.

**Posology and method of administration**

Actrapid® is a fast-acting insulin and may be used in combination with long-acting insulin products.

**Dosage**

Dosage is individual and determined in accordance with the needs of the patient. The individual insulin requirement is usually between 0.3 and 1.0 IU/kg/day. The daily insulin requirement may be higher in patients with insulin resistance (e.g. during puberty or due to obesity) and lower in patients with residual, endogenous insulin production.

An injection should be followed within 30 minutes by a meal or snack containing carbohydrates.

**Dosage adjustment**

Concomitant illness, especially infections and feverish conditions, usually increases the patient's insulin requirement. Concomitant diseases in the kidney, liver or affecting the adrenal, pituitary or thyroid gland can require changes in the insulin dose. Adjustment of dosage may also be necessary if patients change physical activity or their usual diet. Dosage adjustment may be necessary when transferring patients from one insulin preparation to another.

**Administration**

For subcutaneous or intravenous use.

Actrapid® is administered subcutaneously by injection in the abdominal wall. If convenient, the thigh, the gluteal region or the deltoid region may also be used. Injection sites should always be rotated within the same region in order to reduce the risk of lipodystrophy and cutaneous amyloidosis.

Subcutaneous injection into the abdominal wall ensures a faster absorption than from other injection sites.

Injection into a lifted skin fold minimises the risk of unintended intramuscular injection. The needle should be kept under the skin for at least 6 seconds to make sure the entire dose is injected.

Intramuscular administrations are possible under guidance by a physician.

Actrapid® may also be administered intravenously, which should only be carried out by healthcare professionals.

The intravenous use of Actrapid® from any pen or cartridge should be an exception only in situations where vials are not available. In this case, Actrapid® should be drawn into an insulin syringe, provided air is avoided, or infused with an infusion system. This procedure should only be carried out by healthcare professionals.

Actrapid® Penfill® is designed to be used with Novo Nordisk delivery systems and NovoFine® or NovoTwist® needles.

**Contraindications**

Hypersensitivity to the active substance or to any of the excipients.

**Special warnings and precautions for use**

Inadequate dosage or discontinuation of treatment, especially in type 1 diabetes, may lead to **hyperglycaemia**.

Usually, the first symptoms of hyperglycaemia set in gradually, over a period of hours or days. They include thirst, increased frequency of urination, nausea, vomiting, drowsiness, flushed dry skin, dry mouth, loss of appetite as well as acetone odour of breath.

In type 1 diabetes, untreated hyperglycaemic events eventually lead to diabetic ketoacidosis, which is potentially lethal.

**Hypoglycaemia** may occur if the insulin dose is too high in relation to the insulin requirement.

Omission of a meal or unplanned, strenuous physical exercise may lead to hypoglycaemia.

Patients, whose blood glucose control is greatly improved e.g. by intensified insulin therapy, may experience a change in their usual warning symptoms of hypoglycaemia and should be advised accordingly.

Usual warning symptoms may disappear in patients with longstanding diabetes.

Transferring a patient to another type or brand of insulin should be done under strict medical supervision.

Changes in strength, brand (manufacturer), type, origin (human insulin, insulin analogue) and/or method of manufacture may result in a need for a change in dosage. Patients transferred to Actrapid® from another type of insulin may require an increased number of daily injections or change in dosage from that used with their usual insulin products. If an adjustment is needed when switching the patient to Actrapid®, it may occur with the first dose or during the first few weeks or months.

As with any insulin therapy, injection site reactions may occur and include pain, redness, hives, inflammation, bruising, swelling and itching. Continuous rotation of the injection site within a given area may help to reduce or prevent these reactions. Reactions usually resolve in a few days to a few weeks. On rare occasions, injection site reactions may require discontinuation of Actrapid®.

Before travelling between different time zones, the patient should be advised to consult the physician, since this may mean that the patient has to take insulin and meals at different times.

Due to the risk of precipitation in pump catheters, Actrapid® should not be used in insulin pumps for continuous subcutaneous insulin infusion.

### **Skin and subcutaneous tissue disorders**

Patients must be instructed to perform continuous rotation of the injection site to reduce the risk of developing lipodystrophy and cutaneous amyloidosis. There is a potential risk of delayed insulin absorption and worsened glycaemic control following insulin injections at sites with these reactions. A sudden change in the injection site to an unaffected area has been reported to result in hypoglycaemia. Blood glucose monitoring is recommended after the change in the injection site from an affected to an unaffected area, and dose adjustment of antidiabetic medications may be considered.

### **Combination of thiazolidinediones and insulin medicinal products**

Cases of congestive heart failure have been reported when thiazolidinediones were used in combination with insulin, especially in patients with risk factors for development of congestive heart failure. This should be kept in mind if treatment with the combination of thiazolidinediones and insulin medicinal products is considered. If the combination is used, patients should be observed for signs and symptoms of congestive heart failure, weight gain and oedema. Thiazolidinediones should be discontinued if any deterioration in cardiac symptoms occurs.

### **Avoidance of accidental mix-ups/medication errors**

Patients must be instructed to always check the insulin label before each injection to avoid accidental mix-ups between Actrapid® and other insulin products.

### **Interaction with other medicinal products and other forms of interaction**

A number of medicinal products are known to interact with the glucose metabolism.

#### **The following substances may reduce the patient's insulin requirement:**

Oral anti-diabetic products, monoamine oxidase inhibitors (MAOI), non-selective beta-blocking agents, angiotensin converting enzyme (ACE) inhibitors, salicylates, anabolic steroids and sulphonamides.

#### **The following substances may increase the patient's insulin requirement:**

Oral contraceptives, thiazides, glucocorticoids, thyroid hormones, sympathomimetics, growth hormone and danazol.

Beta-blocking agents may mask the symptoms of hypoglycaemia and delay recovery from hypoglycaemia.

Octreotide/lanreotide may either increase or decrease the insulin requirement.

Alcohol may intensify or reduce the hypoglycaemic effect of insulin.

**Pregnancy and lactation**

There are no restrictions on treatment of diabetes with insulin during pregnancy, as insulin does not pass the placental barrier.

Both hypoglycaemia and hyperglycaemia, which can occur in inadequately controlled diabetes therapy, increase the risk of malformations and death *in utero*. Intensified blood glucose control and monitoring of pregnant women with diabetes are recommended throughout pregnancy and when contemplating pregnancy. Insulin requirements usually fall in the first trimester and increase subsequently during the second and third trimesters. After delivery, insulin requirements normally return rapidly to pre-pregnancy values.

There is no restriction on treatment with Actrapid® during breast-feeding.

Insulin treatment of the nursing mother presents no risk to the baby. However, the Actrapid® dosage, diet or both may need to be adjusted.

**Effects on ability to drive and use machines**

The patient's ability to concentrate and react may be impaired as a result of hypoglycaemia. This may constitute a risk in situations where these abilities are of special importance (e.g. driving a car or operating machinery).

Patients should be advised to take precautions to avoid hypoglycaemia while driving. This is particularly important in those who have reduced or absent awareness of the warning signs of hypoglycaemia or have frequent episodes of hypoglycaemia. The advisability of driving should be considered in these circumstances.

**Undesirable effects****a. Summary of the safety profile**

The most frequently reported adverse reaction during treatment is hypoglycaemia. In clinical trials and during marketed use, the frequencies of hypoglycaemia vary with patient population, dose regimens and level of glycaemia control, please see section c below.

At the beginning of the insulin treatment, refraction anomalies, oedema and injection site reactions (pain, redness, hives, inflammation, bruising, swelling and itching at the injection site) may occur. These reactions are usually of transitory nature. Fast improvement in blood glucose control may be associated with acute painful neuropathy, which is usually reversible. Intensification of insulin therapy with abrupt improvement in glycaemic control may be associated with temporary worsening of diabetic retinopathy, while long-term improved glycaemic control decreases the risk of progression of diabetic retinopathy.

**b. Tabulated list of adverse reactions**

Adverse reactions listed below are based on clinical trial data and classified according to MedDRA frequency and System Organ Class. Frequency categories are defined according to the following convention: Very common ( $\geq 1/10$ ); common ( $\geq 1/100$  to  $< 1/10$ ); uncommon ( $\geq 1/1,000$  to  $< 1/100$ ); rare ( $\geq 1/10,000$  to  $< 1/1,000$ ); very rare ( $< 1/10,000$ ); not known (cannot be estimated from the available data).

|                                    |                                                       |
|------------------------------------|-------------------------------------------------------|
| Immune system disorders            | Uncommon – Urticaria, rash                            |
|                                    | Very rare – Anaphylactic reactions*                   |
| Metabolism and nutrition disorders | Very common – Hypoglycaemia*                          |
| Nervous system disorders           | Uncommon – Peripheral neuropathy (painful neuropathy) |
| Eye disorders                      | Uncommon – Refraction disorders                       |

|                                                      |                                     |
|------------------------------------------------------|-------------------------------------|
|                                                      | Very rare – Diabetic retinopathy    |
| Skin and subcutaneous tissue disorders               | Uncommon – Lipodystrophy*           |
|                                                      | Not known – Cutaneous amyloidosis*† |
| General disorders and administration site conditions | Uncommon – Injection site reactions |
|                                                      | Uncommon – Oedema                   |

\* see section c

† ADR from postmarketing sources

### c. Description of selected adverse reactions

#### *Anaphylactic reactions*

The occurrence of generalised hypersensitivity reactions (including generalised skin rash, itching, sweating, gastrointestinal upset, angioneurotic oedema, difficulties in breathing, palpitation, reduction in blood pressure and fainting/loss of consciousness) is very rare but can potentially be life threatening.

#### *Hypoglycaemia*

The most frequently reported adverse reaction is hypoglycaemia. It may occur if the insulin dose is too high in relation to the insulin requirement. Severe hypoglycaemia may lead to unconsciousness and/or convulsions and may result in temporary or permanent impairment of brain function or even death. The symptoms of hypoglycaemia usually occur suddenly. They may include cold sweats, cool pale skin, fatigue, nervousness or tremor, anxiousness, unusual tiredness or weakness, confusion, difficulty in concentration, drowsiness, excessive hunger, vision changes, headache, nausea and palpitation.

#### *Skin and subcutaneous tissue disorders*

Lipodystrophy (including lipohypertrophy, lipoatrophy) and cutaneous amyloidosis may occur at the injection site and delay local insulin absorption. Continuous rotation of the injection site within the given injection area may help to reduce or prevent these reactions.

### **Overdose**

A specific overdose of insulin cannot be defined, however, hypoglycaemia may develop over sequential stages if too high doses relative to the patient's requirements are administered:

- Mild hypoglycaemic episodes can be treated by oral administration of glucose or sugary products. It is therefore recommended that the diabetic patient always carries sugar containing products.
- Severe hypoglycaemic episodes, where the patient has become unconscious, can be treated with glucagon (0.5 to 1 mg) given intramuscularly or subcutaneously by a trained person, or with glucose given intravenously by a healthcare professional. Glucose must be given intravenously, if the patient does not respond to glucagon within 10 to 15 minutes.

Upon regaining consciousness, administration of an oral carbohydrate is recommended for the patient in order to prevent a relapse.

### **Pharmacodynamic properties**

Pharmacotherapeutic group: Drugs used in diabetes. Insulins and analogues for injection, fast-acting, insulin (human). ATC code: A10AB01.

The blood glucose lowering effect of insulin is due to the facilitated uptake of glucose following binding of insulin to receptors on muscle and fat cells and to the simultaneous inhibition of glucose output from the liver.

A clinical trial in a single intensive care unit treating hyperglycaemia (blood glucose above 10 mmol/L) in 204 diabetic and 1344 non-diabetic patients undergoing major surgery showed that normoglycaemia (blood glucose 4.4 – 6.1 mmol/L) induced by intravenous Actrapid® reduced mortality by 42% (8% versus 4.6%). Actrapid® is a fast-acting insulin.

Onset of action is within ½ hour, reaches a maximum effect within 1.5 – 3.5 hours and the entire duration of action is approximately 7 - 8 hours.

### **Pharmacokinetic properties**

Insulin in the blood stream has a half-life of a few minutes. Consequently, the time-action profile of an

insulin preparation is determined solely by its absorption characteristics.

This process is influenced by several factors (e.g. insulin dosage, injection route and site, thickness of subcutaneous fat, type of diabetes). The pharmacokinetics of insulin products are therefore affected by significant intra- and inter-individual variation.

### **Absorption**

The maximum plasma concentration is reached within 1.5 – 2.5 hours after subcutaneous administration.

### **Distribution**

No profound binding to plasma proteins, except circulating insulin antibodies (if present) has been observed.

### **Metabolism**

Human insulin is reported to be degraded by insulin protease or insulin-degrading enzymes and possibly protein disulfide isomerase. A number of cleavage (hydrolysis) sites on the human insulin molecule have been proposed; none of the metabolites formed following the cleavage are active.

### **Elimination**

The terminal half-life is determined by the rate of absorption from the subcutaneous tissue. The terminal half-life ( $t_{1/2}$ ) is therefore a measure of the absorption rather than of the elimination *per se* of insulin from plasma (insulin in the blood stream has a  $t_{1/2}$  of a few minutes). Trials have indicated a  $t_{1/2}$  of about 2 - 5 hours.

### **Children and adolescents**

The pharmacokinetic profile of Actrapid® has been studied in a small number (n=18) of diabetic children (aged 6 - 12 years) and adolescents (aged 13 - 17 years). The data are limited but suggest that the pharmacokinetic profile in children and adolescents may be similar to that in adults.

However, there were differences between age groups in  $C_{max}$ , stressing the importance of individual dose titration.

### **Preclinical safety data**

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, toxicity to reproduction.

### **List of excipients**

Zinc chloride, glycerol, metacresol, sodium hydroxide/hydrochloric acid (for pH adjustment) and water for injections.

### **Incompatibilities**

Insulin products should only be added to compounds with which it is known to be compatible. Medicinal products added to the insulin solution may cause degradation of the insulin, e.g. if the medicinal products contain thiols or sulphites.

### **Special precautions for storage**

#### **Unopened**

Store in a refrigerator (2°C - 8°C). Keep away from the cooling element. Do not freeze.

If there are 6 months or more until the expiry date, the product may be stored outside of the refrigerator (below 30°C) for a maximum of 4 weeks before it is taken into use or carried as a spare. After storage outside of the refrigerator, the product must not be returned to the refrigerator. Please record the beginning of storage outside of the refrigerator on the product carton.

#### **Opened**

During use or when carried as a spare: Do not refrigerate or freeze.

The in-use shelf life is 6 weeks when stored below 30°C.

### **General Information**

Keep the cartridge in the outer carton in order to protect from light.

Actrapid® must be protected from excessive heat and light.

### **Nature and contents of container**

3 ml solution in cartridge (type 1 glass) with a plunger (bromobutyl) and a stopper (bromobutyl/polyisoprene) in a carton.

Pack sizes of 1, 5 and 10 cartridges.

Not all pack sizes may be marketed.

**Special precautions for disposal and other handling**

For intravenous use, infusion systems with Actrapid® at concentrations from 0.05 IU/ml to 1.0 IU/ml insulin human in the infusion fluids: 0.9% sodium chloride, 5% dextrose and 10% dextrose with 40 mmol/L potassium chloride, using polypropylene infusion bags, are stable at room temperature for 24 hours.

Although stable over time, a certain amount of insulin will initially be absorbed to the material of the infusion bag. Monitoring of blood glucose is necessary during the infusion.

Cartridges should only be used in combination with products that are compatible with them and allow the cartridge to function safely and effectively.

Needles and Actrapid® Penfill® must not be shared. The container must not be refilled.

Insulin preparations which have been frozen must not be used.

Insulin solutions should not be used if they do not appear clear and colourless.

Actrapid® should not be used in insulin pumps for continuous subcutaneous insulin infusion. The patient should be advised to discard the needle after each injection.

In case of emergency in current Actrapid® users (hospitalisation or insulin pen malfunction), Actrapid® can be withdrawn with an U100 insulin syringe from the cartridge.

**Manufactured by:**

Novo Nordisk Produção Farmacêutica do Brasil Ltda.

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